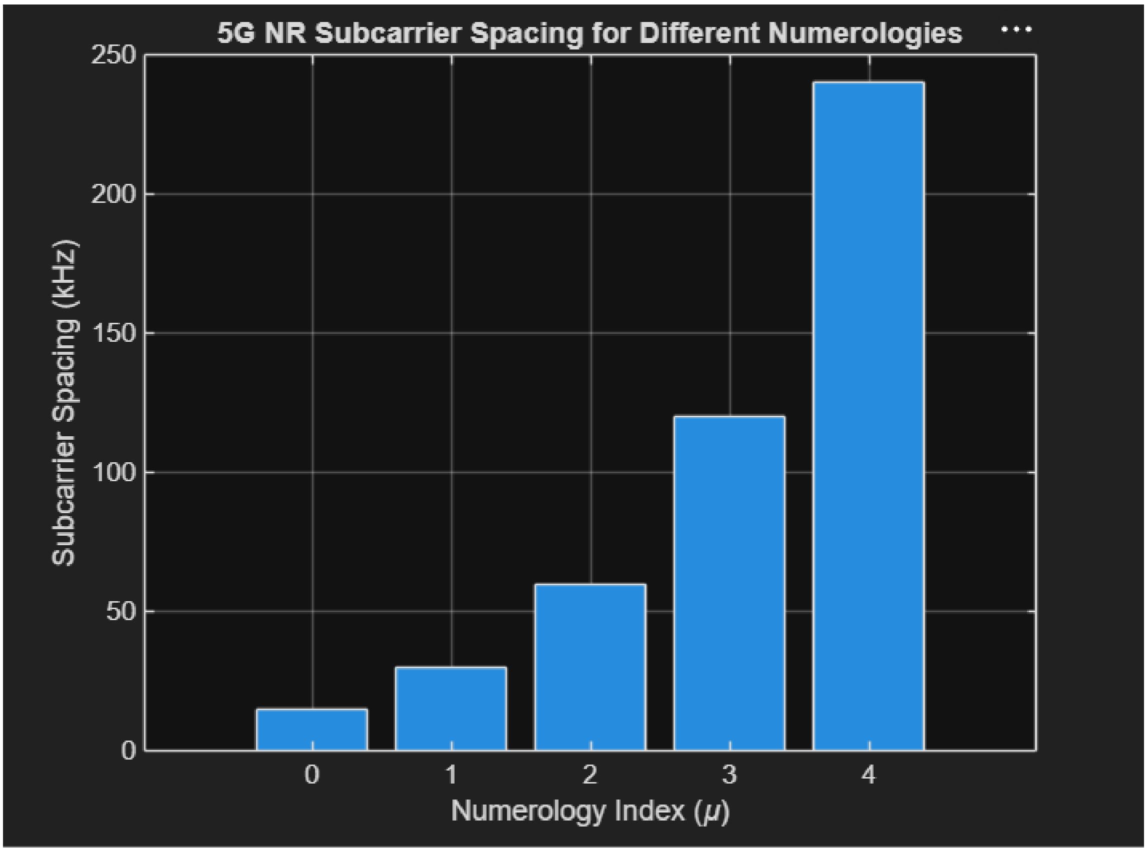
Experiment: 6

5G NR Frame Structure Analysis with Different Subcarrier Spacing

OBJECTIVE 1:

CODE:

```
clc;
clear;
close all;
scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)
mu = 0:4; % Numerology Index
bar(scs)
grid on
set(gca,'XTick',1:5)
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```

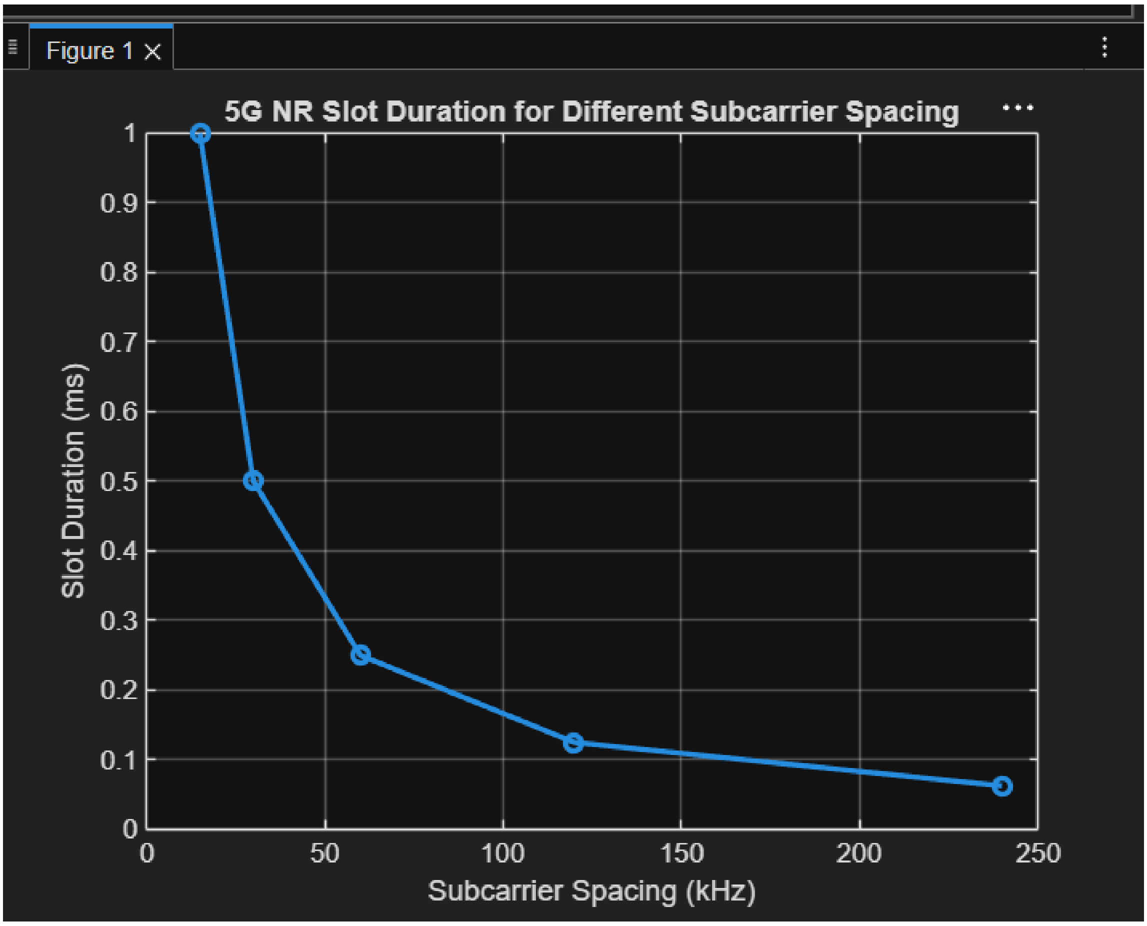


OBJECTIVE 2:

CODE:

```
clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz) slot = [1
0.5 0.25

0.125 0.0625]; % Slot Duration (ms)

plot(scs, slot, 'o-', 'LineWidth', 2)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Slot Duration (ms)')
```

title('5G NR Slot Duration for Different Subcarrier Spacing')



OBJECTIVE 3:

CODE:

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

symbols = [14 14 14 14 14]; % OFDM Symbols per Slot

figure

bar(scs, symbols)

grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Number of OFDM Symbols')

```
title('OFDM Symbols per Slot for Different Subcarrier Spacing')

ylim([0 16])
```